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Circadian control of a sex-specific behaviour in *Drosophila*

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eLife Assessment

This **important** study introduces an experimental approach for studying *Drosophila* oviposition rhythms and identifies the subset of circadian clock neurons that mediate the circadian control of oviposition. The authors try to resolve a known noisy rhythm and provide **convincing** evidence by using statistical averaging techniques which help reduce this noise but at the cost of variation across individual rhythms. To this end, including the time series of representative individuals for all genotypes tested would have helped in interpreting some of the results. This paper will be of interest to anyone interested in insect ovarian physiology, circadian biology, and reproductive fitness.

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Abstract

An endogenous circadian clock controls many of the behavioral traits of *Drosophila melanogaster*. This “clock” relies on the activity of interconnected clusters of neurons that harbor the clock machinery. The hierarchy among clusters involved in the control of rest-activity cycles has been extensively studied. Sexually dimorphic behaviors, on the other hand, have received less attention. Even though egg-laying, a female characteristic behavior, has been shown to be rhythmic, it remains largely unexplored possibly due to methodological constraints. The current study provides the first steps towards determining the neural substrates underlying the circadian control of egg-laying. We show that, whereas the lateral ventral neurons (LN_{vs}) and the dorsal neurons (DN_s) are dispensable, the lateral dorsal neurons (LN_{ds}) are necessary for rhythmic egg-laying. Systematically probing the *Drosophila* connectome for contacts between circadian clusters and oviposition-related neurons, we found no evidence of direct connections between LN_{vs} or DN_s and neurons recruited during oviposition. Conversely, we did find bidirectional connections between Cryptochrome (Cry) expressing LN_d (Cry+ LN_ds) and oviposition related neurons. Taken together, these results reveal that Cry+ LN_d neurons have a leading role in the control of the egg-laying rhythm in *Drosophila* females.

Introduction

Most organisms are capable of coordinating their physiology and behavior with the 24 hours of day/night cycling generated by the Earth’s rotation. These biological rhythms are driven by molecular clocks that are conserved across animals (Allada et al., 2001 [↗](#)). In *Drosophila melanogaster* the core clock components comprise a transcription/translation feedback loop with four core proteins: CLOCK (CLK) CYCLE (CYC), PERIOD (PER), and TIMELESS (TIM) (Hardin, 2005 [↗](#); Tataroglu and Emery, 2015 [↗](#)). In brief, CLK and CYC activate transcription of *period* and *timeless* genes, which, once translated into PER and TIM proteins, dimerize and translocate into the nucleus where they bind to CLK and CYC, thereby inhibiting their own transcription. This molecular feedback loop has a period of approximately 24 hours. In the *Drosophila* brain this

molecular circadian clock is expressed in ~ 242 neurons, which are organized in different clusters based in gene expression, anatomy and localization (Helfrich-Förster, 2005a [↗](#); Reinhard et al., 2024 [↗](#); Yoshii et al., 2012 [↗](#)). These clusters are: ventrolateral neurons (LNV; encompassing the small and large LNV groups), dorsolateral neurons (LND), lateral posterior (LPN), and dorsal neurons [DN; separated in DN1, 2, and 3 and further subdivided into anterior (DN1a) and posterior (DN1p) DN1]. Rhythmic locomotor behavior depends primarily on the activity of the sLNVs, LNDs, and DN1s, but all clusters contribute to different extents (Helfrich-Förster, 2005b [↗](#)). Within the lateral neuron group, the sLNV are important because they drive locomotor rhythmicity under free-running (DD) conditions (Grima et al., 2004 [↗](#); Stoleru et al., 2004 [↗](#)), through the release of Pigment Dispersing Factor (PDF), a neuropeptide relevant for communication between clock neurons (Renn et al., 1999 [↗](#); Yoshii et al., 2009 [↗](#)).

Egg-laying is one of the most important female behaviors since it has a profound impact on the fitness of a species. Egg laying is largely governed by successful mating, but is also influenced by circadian and seasonal rhythms and by environmental factors such as temperature and food availability, among others (Cury et al., 2019 [↗](#)). The circadian rhythm of oviposition is one of the less studied rhythms in *Drosophila*, possibly due to the challenges involved in monitoring and recording this behavior. The periodic deposition of eggs involves a series of events ranging from the production of oocytes to the choice of the most appropriate substrate for the eggs (Allemand, 1976 [↗](#); Yang et al., 2008 [↗](#)). The circadian nature of this behavior was revealed by its persistence under DD with a period around 24 h and a peak near night onset (McCabe and Birley, 1998a [↗](#); Sheeba et al., 2001a [↗](#)). Egg-laying rhythmicity is temperature and nutrition compensated. (Howlader et al., 2006a [↗](#)). Moreover, oviposition is rhythmic in virgin females as well as in mated ones suggesting that this rhythm is not regulated by the act of mating and, instead, is endogenously driven (Menon et al., 2014a [↗](#)).

Although oviposition exhibits a circadian component, the molecular and neural substrates that govern this rhythm have only been partially described (Howlader et al., 2006b [↗](#)). Here, using an automated egg collection device developed in our lab, we examined the contribution of the molecular clock in specific neuronal clusters. By downregulating *per* in subsets of circadian neurons, we determined that even though impairing the molecular clock in the entire clock network reduced the power of the circadian rhythm of oviposition, restricting the disruption to DN1ps or LNVs did not (confirming results obtained by Howlader et al., 2006 for the sLNV (Howlader et al., 2006b [↗](#))). Interestingly, egg-laying rhythms disappeared upon targeting a subset of LND clocks (Cry+ LND) through RNAi-mediated *per* downregulation, a condition where rhythmicity of locomotor activity patterns remained unaltered, suggesting a leading and very specific role of LND clocks/neurons in the control of oviposition. Finally, the assessment of the synaptic connectivity between clock and oviposition controlling neurons using the Janelia hemibrain dataset revealed direct synaptic connections between this subset of LNDs and oviposition neurons, which is consistent with the essential role of LNDs neurons in the control of this behavior.

Results

Egg-laying is rhythmic when registered with a semiautomated egg collection device

Oviposition in *Drosophila* is one of the less studied behaviors regulated by the circadian clock. This is probably due to the difficulties involved in monitoring and recording it, and the current lack of standard devices to accomplish this. Egg collection and counting is usually done manually, making the experiments particularly demanding and labor-intensive. In this approach, eggs are typically collected every 4 hours (sometimes also every 2 hours), which usually implies transferring the fly to a new vial or extracting the food with the eggs and replacing it with fresh food in the same vial (Menon et al., 2014b [↗](#)). Either way this implies disturbing the fly several times a day which could alter the normal rhythm of oviposition, and demands the intervention of an experimenter every 4 hours during several days. In order to avoid this, we developed a semiautomated egg collection

device where 21 flies are individually housed, and each enclosure is slowly shifted every 4 hours from one food patch to a new food patch. Once per day all food patches are collected and eggs are manually counted (see **Material and Methods** and [Figure supplement 1](#) [\[1\]](#) for more details).

The egg records of fruit flies display several distinct features that complicate the assessment of rhythmicity. The recorded variable is discrete and typically takes only a few values, depending on the sampling frequency. The more often it is sampled, the fewer number of values the variable takes. Thus, it is only possible to sample it a few times per cycle (usually less than ten). For noisy signals, this feature makes the assessment of rhythmicity much more complicated, compared with signals that have similar noise, but are sampled more frequently (as is the case for activity records). The particularities of egg-laying in *Drosophila* add further complications. First, the number of eggs laid shows a clear decreasing trend over time. Second, this behavior appears to be particularly “noisy”, in the sense that the decision to lay an egg depends not only on the time of the day, but also on many other stimuli ([Cury et al., 2019](#) [\[2\]](#)). As a consequence, any method that tries to assess the presence of a rhythm must be able to subtract, at least partially, the effects of trend and noise.

In order to achieve this, we have considered the register of the eggs laid by a “rhythmic” female as a periodic signal corrupted by random noise (whereas for arrhythmic females we assume the record is completely random). It is well known that light is a powerful zeitgeber for activity, synchronizing all the flies in a population to the light-dark (LD) cycle, and that even after lights off, activity remains synchronized between flies for quite a long time. Since this is caused by the synchronization of cellular clocks, we assume that a similar effect occurs for oviposition. Thus, under LD conditions, or in the first days of DD, it can be assumed that the periodic signal is the same for all flies, whereas the noisy component is different for each individual. In this case it is well known (see e.g. ([van Dronghen, 2007](#) [\[3\]](#))) that averaging of several “replicates” (individual flies, in our case) leads to an improvement of the signal-to-noise ratio allowing the emergence of the underlying rhythm. As this approach seems to be better suited for detecting noisy rhythms, in the rest of the paper the assessment of rhythmicity will be based in the record of eggs laid during a given period of time averaged over all females of each genotype. [Figure 1](#) [\[4\]](#) shows that, when eggs are collected with our device in LD, *Canton-S* females have a peak of egg deposition at the end of the day, as previously described in the literature ([Bailly et al., 2023](#) [\[5\]](#); [Howlader et al., 2006a](#) [\[6\]](#); [Sheeba et al., 2001b](#) [\[7\]](#)) ([Figure 1A](#) [\[4\]](#)). [Figure 1E](#) [\[4\]](#) shows that only about 70% of individuals are rhythmic (with a period close to 24 h [Figure 1F](#) [\[4\]](#)) although the average egg-laying profile is robust with a period around 24 h ([Figure 1B](#) [\[4\]](#)). Under DD conditions, the percentages of rhythmic individuals are the same as in LD ([Figure 1E](#) [\[4\]](#)) and the average profile also shows a robust rhythm with a period of 24 h ([Figure 1D](#) [\[4\]](#)). It is well known that in locomotor activity experiments more than 90% of the flies are rhythmic in DD, and virtually all flies are rhythmic in LD (see e.g. [Shindey et al. 2016](#), and [Deppisch et al. 2022](#) [\[8\]](#) ([Deppisch et al., 2022](#) [\[9\]](#); [Shindey et al., 2017](#) [\[10\]](#))). This shows that, at the individual level, egg-laying is much less robust than locomotor activity, compounding the difficulty of observing the influence of the circadian clock on this behavior.

In order to check that our egg collection device does not in any way entrain egg laying in flies, we tested it with *perS* mutants ([Konopka and Benzer, 1971](#) [\[11\]](#)). These flies are known to display a short period of locomotor activity, both in males ([Konopka and Benzer, 1971](#) [\[11\]](#)) and in females ([McCabe and Birley, 1998b](#) [\[12\]](#)). Additionally, it has been shown that oviposition in *perS* flies is rhythmic, with a period of 20.6±0.9h ([McCabe and Birley, 1998b](#) [\[12\]](#)). Our results for this mutant are in agreement with this ([Figure 1J](#) [\[4\]](#) shows a period of the average at 18.8±1.1 h), thus confirming that flies are not entrained by our egg collection device. It is important to notice that in the *yellow white* (YW) control strain, the individual rhythmicity was rather low (around 50% [Figure 1K](#) [\[4\]](#)), but the average is strongly rhythmic ([Figure 1H](#) [\[4\]](#)) thus confirming the need for an analysis method that can take into account the low signal-to-noise-ratio of this behavior.

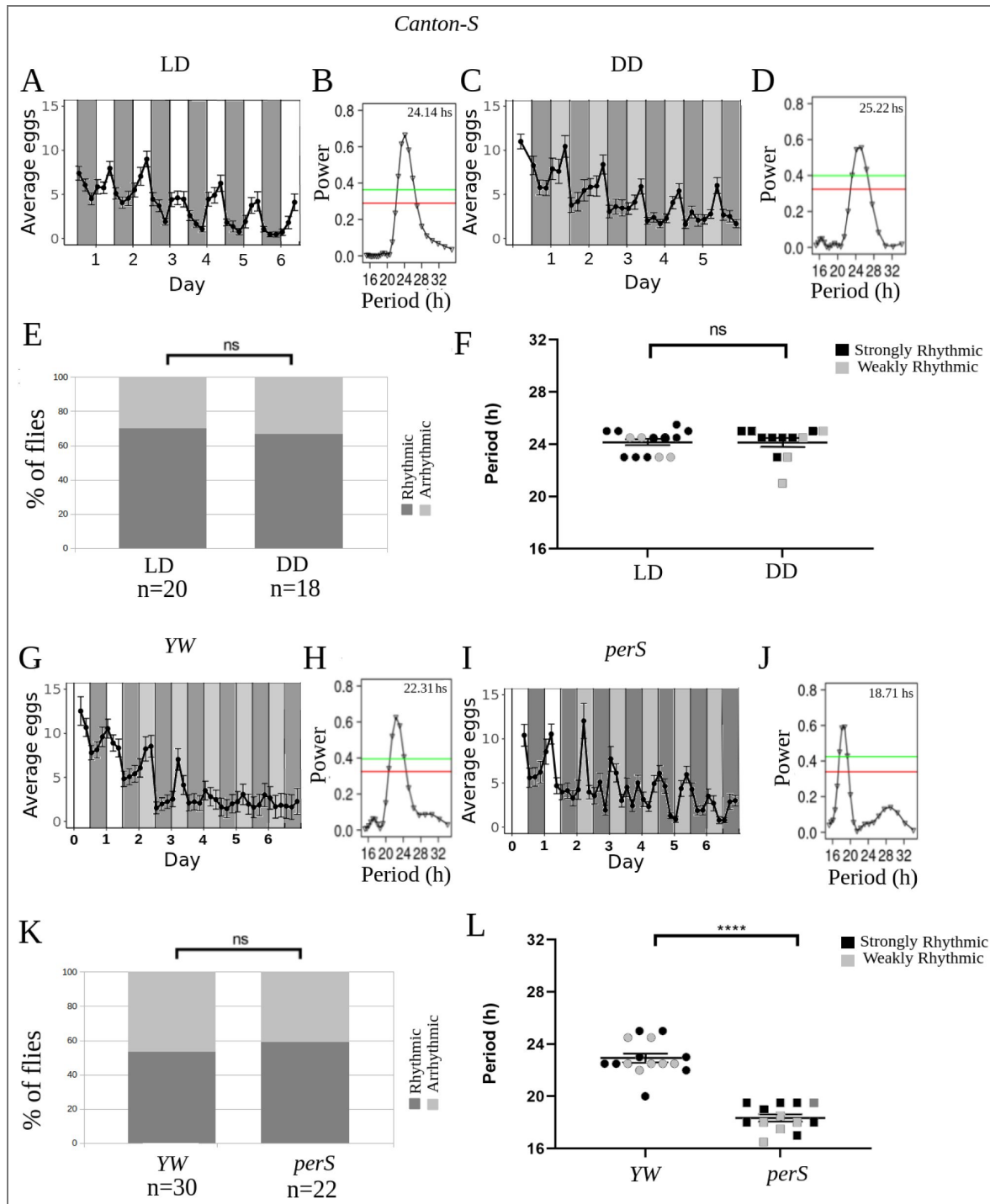


Figure 1. Oviposition in *Drosophila* is rhythmic when registered with our semiautomated egg collection device.

A, C, G, I: Average eggs collected as a function of time in four different experiment (*Canton-S*, LD: **A**, *Canton-S*, DD: **C**; *yw*: **G**; *perS*: **I**). The average was made with all flies (Strongly rhythmic, weakly rhythmic and arrhythmic). White and dark grey bars represent periods of lights on and off respectively (LD), whereas light gray bars represent subjective days, DD, (i.e times where lights were on at rearing, but are now off). **B, D, H, J:** Lomb-Scargle periodograms of all genotypes, made with all flies, strongly rhythmic, weakly rhythmic and arrhythmic. Red and green horizontal lines represent significances of 0.05 and 0.01, respectively. **E, K:** Percentage of females with rhythmic oviposition (**E**: *Canton-S* in LD and DD, **K**: *yw* and *perS*). The rhythmic flies include both strongly and weakly rhythmic flies. **F, L:** Period of oviposition rhythms for rhythmic individual flies (**F**: *Canton-S* in LD and DD, **L**: *yw* and *perS*). ns: non significant, ****p<0.001 (chi-squared test for scatter plots and proportions).

Downregulation of *per* in clock neurons causes a dramatic reduction in the power of egg-laying rhythms

The circadian system of *Drosophila* comprises a central clock located in the brain that controls locomotor behavior, as well as peripheral clocks located in many tissues, that regulate distinct behaviors (Franco et al., 2018 [↗](#); Ito and Tomioka, 2016 [↗](#)). For example, olfaction rhythms are controlled by peripheral clocks in the antennae, and eclosion rhythms are partially controlled by peripheral clocks in the prothoracic gland (Myers et al., 2003 [↗](#); Tanoue et al., 2004 [↗](#)). Earlier work proposed that oviposition might also be controlled by a peripheral clock (Manjunatha et al., 2008 [↗](#)). In order to test for this possibility, we used RNA interference (RNAi) to knock down the clock protein PER (Herrero et al., 2017 [↗](#); Zhang et al., 2021 [↗](#)) in all clock neurons in the brain (using the *Clk856* driver (Gummadova et al., 2009 [↗](#))), thereby disrupting their molecular clock. First, we tested the effectiveness of this tool by monitoring the locomotor activity of males with their endogenous clock silenced in all clock neurons (*Clk856-Gal4>UAS-per^{RNAi}*). Supplementary Table 1 [↗](#) shows that these flies exhibited a significant decrease in locomotor rhythmicity compared to their genetic controls. Having confirmed the effectiveness of the RNA interference, we then monitored egg-laying in females from the previously mentioned lines.

As shown in Figure 2D, E [↗](#), when PER is knocked down in all clock neurons oviposition rhythmicity is drastically reduced compared to controls Figure 2B, C [↗](#). Notice, however, that the peak observed in Fig. 2C [↗](#) has not disappeared completely in 2E, and is in fact not far from the significance threshold. This could be due to some remaining rhythmicity in some individuals (although noise precludes the observation of this at the individual level). Interestingly, when locomotor activity is studied in male of this genotype, there is a fraction (less than 40%) of individuals that remain rhythmic in DD (see Supplementary Table 1 [↗](#)).

This could be due either to the contribution of some peripheral oscillator/s relevant for the control of this behavior, or the presence of some residual PER in clock cells (Herrero et al., 2017 [↗](#)).

LNv and DN1 neurons are not necessary for egg-laying rhythmicity

We next sought to establish the role of the most important circadian groups in the control of oviposition. We concentrated on three groups: LNv, DN1p and LNd neurons. The LNvs are known to be essential for the maintenance of circadian locomotor activity rhythms, and are the only ones that express the PDF neuropeptide (Renn et al., 1999 [↗](#); Yoshii et al., 2009 [↗](#)). However, we found that *per^{RNAi}* mediated disruption of the molecular clock in PDF+ cells does not abolish the time-of-day dependent oviposition (see Figure 2G,H,I,J [↗](#)). This is in line with previous results describing rhythmic oviposition after ablation of all PDF+ neurons (Howlader et al., 2006a [↗](#)). In addition, we observed a shortening of the egg-laying period when compared to controls, (20.74 h $\pm$ 1.21 h vs 24.35 h $\pm$ 1.83 h respectively, Figure 2H,J [↗](#)). This shortening can also be observed at the individual level (Supplementary Figure 2S1 [↗](#)). Taken together, these results suggest that these neurons may have some influence on the control of this behavior.

The DN1 neurons are not essential for the maintenance of locomotor rhythms in DD, but they contribute to the siesta (Guo et al., 2016 [↗](#)) and generation of evening activity (Helfrich-Förster et al., 2007 [↗](#); Zhang et al., 2010 [↗](#)) when the peak of egg deposition occurs. Interestingly, it has recently been shown that a posterior subset (DN1p) drives the circadian rhythm of oogenesis through the neuropeptide allatostatin C (Zhang et al., 2021 [↗](#)). To examine whether the molecular clock of DN1p neurons is involved in the control of rhythmic oviposition, we evaluated the impact of PER downregulation on mated females expressing the *Clk4.1* driver (which is expressed in ~10 DN1ps per hemisphere (Zhang et al., 2010 [↗](#))). As Figure 2L, M, N, O [↗](#) shows, the disruption of the clock in these neurons does not alter the rhythmicity of egg-laying in DD. To further confirm these results, and in the process examine if some other features of the DN1p could contribute to the control of oviposition, we silenced these neurons through the expression of the inward rectifier potassium channel *kir2.1* (Baines et al., 2001 [↗](#)). Silencing DN1p neurons did not affect the

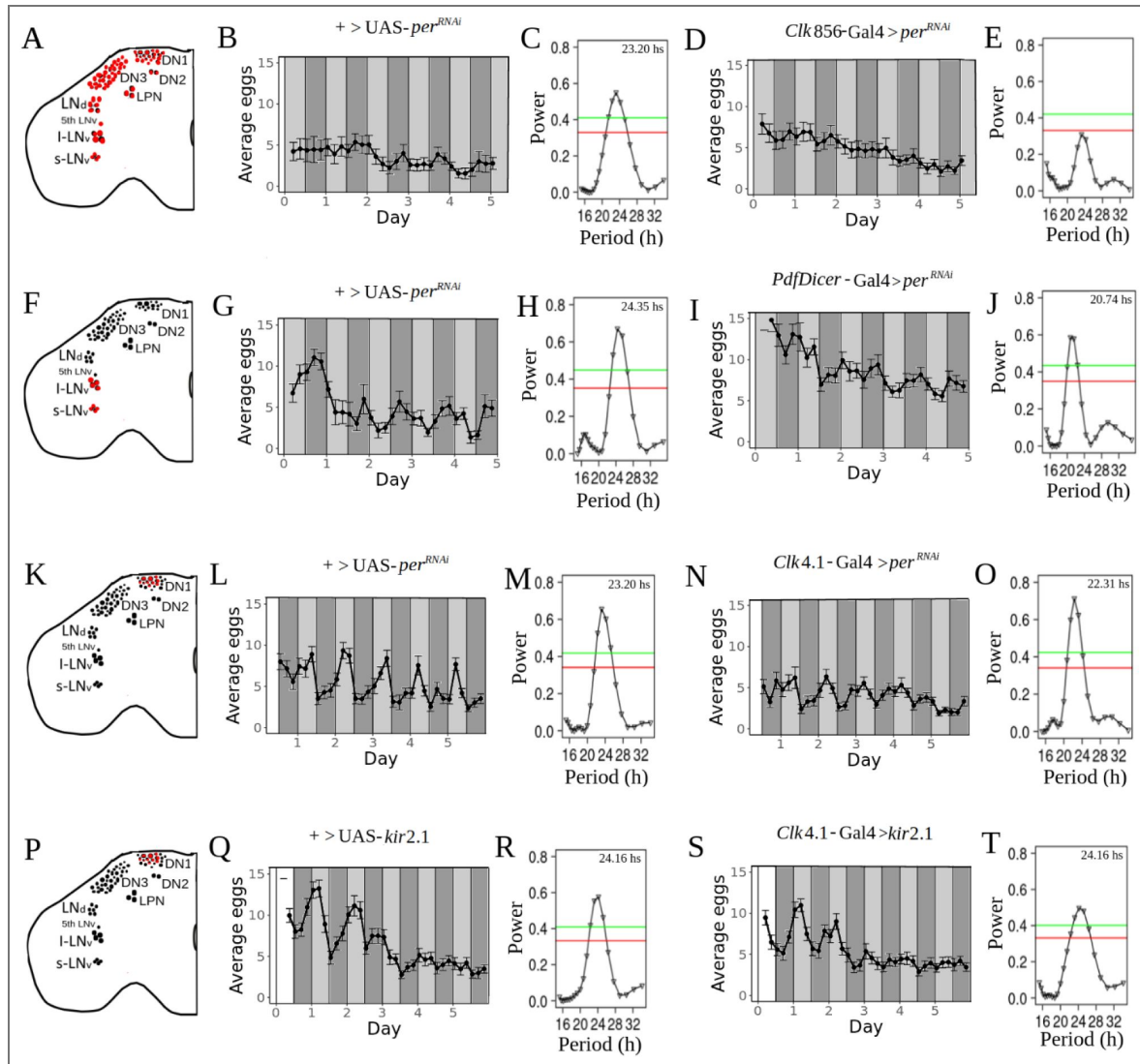


Figure 2. Circadian rhythmicity of oviposition is dramatically reduced when the molecular clock is disrupted in all clock neurons, but not when only LNV or DN1 neurons are affected.

A, F, K, P: Schematic diagram of the neurons (painted in red) where the molecular clock has been disrupted. **B, D, G, I, L, N, Q, S:** Average eggs collected as a function of time. The average was made with all flies (strongly rhythmic, weakly rhythmic and arrhythmic). White and dark grey bars represent periods of lights on and off respectively (LD), whereas light gray bars represent subjective days, DD, (i.e. times where lights were on at rearing, but are now off). **C, E, H, J, M, O, R, T:** Lomb-Scargle periodograms of all genotypes (made with all flies, rhythmic, weakly rhythmic and arrhythmic). Red and green horizontal lines represent significances of 0.05 and 0.01, respectively. **B, C:** $+>UAS-per^{RNAi}$, $n=22$. **D, E:** $Clk856-Gal4>per^{RNAi}$, $n=26$. **G, H:** $+>UAS-per^{RNAi}$, $n=18$. **I, J:** $PdfDicer-Gal4>per^{RNAi}$, $n=34$. **L, M:** $+>UAS-per^{RNAi}$, $n=38$. **N, O:** $Clk4.1-Gal4>per^{RNAi}$, $n=40$. **Q, R:** $+>UAS-kir2.1$, $n=36$. **S, T:** $Clk4.1-Gal4>UAS-kir2.1$, $n=35$.

rhythmicity of egg-laying in females which showed similar patterns to those exhibited by controls females (Figures 2Q, R, S, T [↗](#)). Taken together, these results show that neither LNV nor DN1p neurons play an important role for rhythmic oviposition.

The molecular clock in Cry+ LNd neurons is necessary for rhythmic egg-laying

The LNd neurons include 6 cells whose expression pattern is very heterogeneous (Hermann-Luibl and Helfrich-Förster, 2015 [↗](#); Ma et al., 2021 [↗](#)). This group contains a subset of 3 Cry+ LNd neurons that express both *cry* (which encodes the light sensing protein cryptochrome) and *pdf* (the PDF receptor), and 3 Cry- LNd neurons, that express neither of these genes (Im et al., 2011 [↗](#); Yoshii et al., 2008 [↗](#)). Even though the LNds are not essential for rhythmic locomotor activity under constant conditions the Cry+ LNds, together with the 5-th sLNV, termed the “E oscillator”, are the main drivers of evening activity (Stoleru et al., 2004 [↗](#)). These two LNd groups greatly differ in their connectivity to the rest of the brain, since Cry+ LNds have both many more outputs and inputs than the Cry- LNds (Shafer et al., 2022 [↗](#)). In order to assess the role of the Cry+ LNds in the control of oviposition, we employed a Gal4 driver, *Mai179-Gal4*; *pdf-Gal80* (Grima et al., 2004 [↗](#); Picot et al., 2007 [↗](#)), to direct expression to E cells. This driver is expressed in the 5th sLNV, 3 Cry+ LNds, and also drives weak expression in a small subset of DN1s. As Figure 3 G [↗](#) shows, impairing clock function in Cry+ LNds results in a drastic reduction of rhythmic egg-laying compared with the genetic controls (Figure 3 C, E [↗](#)). To further confirm this observation, we used another approach to disrupt the molecular clock in those neurons, resorting to the very specific MB122B driver (Duhart et al., 2020 [↗](#); Guo et al., 2017 [↗](#)). We also used a different UAS sequence, a dominant negative version of the clock gene *cycle* (Balaji Krishnan, 1999 [↗](#)). As expected, under these conditions rhythmic egg laying was also severely compromised (Figure 3M, N [↗](#)), as compared with the genetic controls (Figure 3I, J, K, L [↗](#)). Moreover, when we used RNAi to knock down the clock protein PER with the MB122B driver, the results were very similar to those obtained with the *Mai179-Gal4*; *pdf-Gal80* driver (Supplementary figure 3S1 [↗](#)). Thus, we conclude that the molecular clock in Cry+ LNd neurons is necessary for the generation of egg-laying rhythms.

The appearance of two non-significant peaks in Figure 3 G [↗](#) (and Supp. Fig 3S1 [↗](#)) might suggest that these genetic manipulations of the LNds neurons generate weakened complex rhythms (Beckwith and Ceriani, 2015 [↗](#)) instead of complete arrhythmicity. This would mean that, at the individual level, there should be either many flies with a corresponding complex rhythm, or two weakly rhythmic subpopulations, each one accounting for one of the peaks of the average periodogram. However, an examination of the individual periodograms (Supplementary Figures 3S2 and 3S3 [↗](#)) does not provide convincing evidence to support any of these two hypotheses. On the other hand, when the method is applied to synthetic, completely random time series, it can generate periodograms with two non significant peaks (as shown in panels 2,3, and 4 of Figure S12 [↗](#) of the Supplementary Material). Thus, we favor the hypothesis that the two peaks present in Figure 3 G [↗](#) (and Supplementary Figure 3S1 [↗](#)) are spurious.

It has recently been shown that disrupting the molecular clock in E cells by means of the MB122B driver did not alter locomotor activity rhythms (Bulthuis et al., 2019 [↗](#)). This would imply that locomotor activity and oviposition rhythms are controlled by distinct groups of clock neurons. However, Bulthuis et al. (2019) [↗](#) only assessed the locomotor activity of male flies, and it is well known that, unlike males, females undergo important changes in their activity after mating, such as the loss of the midday siesta (Elwyn Isaac et al., 2010 [↗](#)) or the morning anticipation (Riva et al., 2022 [↗](#)). For these reasons, we decided to examine the influence of the molecular clock of E cells directly on the locomotor activity of mated females. The standard setup for assessing locomotor activity is the *Drosophila* Activity Monitoring (DAM) system (Pfeifferberger et al., 2010 [↗](#)), where flies are housed in small glass tubes (of 5mm in diameter). Such arena does not offer an adequate environment for testing flies that lay eggs during several days; instead, we monitored activity using an alternative system developed in our lab, where flies are housed in much larger cubicles and their activity is recorded by a video system (see Methods and (Riva et al., 2022 [↗](#))).

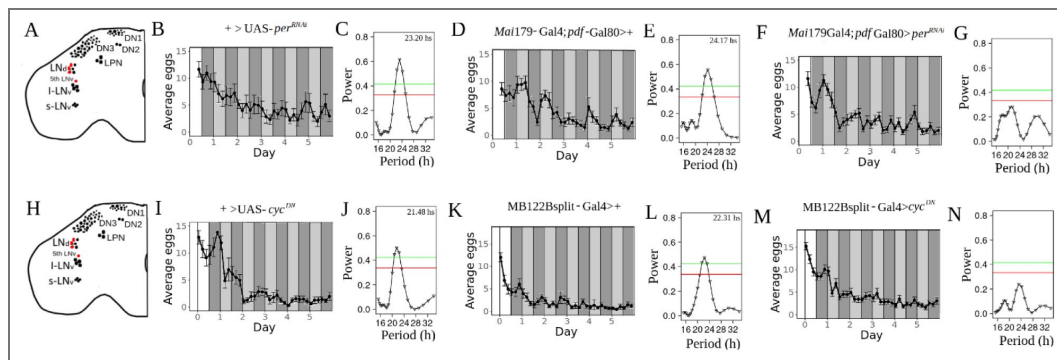


Figure 3. Disruption of the molecular clock in Cry+ LNd neurons drastically reduces the circadian rhythmicity of oviposition.

A, H: Schematic diagram of the neurons (painted in red) where the molecular clocks have been disrupted. **B, D, F, I, K, M:** Average eggs collected as a function of time. The average was made with all flies (Strongly rhythmic, weakly rhythmic and arrhythmic). White and dark grey bars represent periods of lights on and off respectively (LD), whereas light gray bars represent subjective days, DD, (i.e times where lights were on at rearing, but are now off). **C, E, G, J, L, N:** Lomb-Scargle periodograms of all genotypes (made with all flies, rhythmic, weakly rhythmic and arrhythmic). Red and green horizontal lines represent significances of 0.05 and 0.01, respectively. **B, C:** +>UAS-*per*^{RNAi}, n= 15. **D, E:** *Mai179-Gal4; pdf-Gal80>+*, n= 17. **F, G:** *Mai179-Gal4; pdf-Gal80>UAS-per*^{RNAi}, n=28. **I, J:** +>UAS-*cyc*^{DN}, n=15. **K, L:** *MB122Bsplit-Gal4>+*, n=14. **M, N:** *MB122Bsplit-Gal4>UAS-cyc*^{DN}, n=30.

Interestingly, mated females with disrupted E cell clocks were as rhythmic as control flies (Figure 4), showing that there is no sex-specific contribution of this cluster to the control of rhythmic rest-activity cycles, which opens the possibility that E cells differently contribute to the control of oviposition and locomotor activity. In other words, our results show that these two important rhythmic behaviors are primarily controlled by distinct groups of clock neurons.

Cry+ LNd neurons directly contact oviIN

If the circadian clock influences oviposition, at least some clock neurons should be connected, directly or indirectly, to the neurons in the brain that control the motor program leading to egg deposition. These neurons, that were recently characterized (Vijayan et al., 2023a; Wang et al., 2020; Zhou et al., 2014), include: the oviposition descending neurons (oviDN, 5 per hemisphere), the oviposition excitatory neurons (oviEN, 1 per hemisphere), the oviposition inhibitory neurons (oviIN, 1 per hemisphere), and pC1 neurons (5 per hemisphere). The best tool in hand to systematically probe the connectivity between circadian clock and oviposition-related neurons is the *Drosophila* hemibrain connectome (Scheffer et al., 2020). This is the result of the FlyEM connectomics project, which used electron microscopy images of the brain of a 5-day old female fly to reconstruct the connectivity map (or connectome) of ~25000 neurons, mostly from the central part and the right lateral part of the brain.

28 of these neurons have been identified as clock neurons, and are located in the right part of the brain (except for one LPN (Scheffer et al., 2020)). These include the best characterized circadian groups (sLNvs, lLNvs, LNdS). For other groups, such as the DN1s and LPNs, only a minor fraction have been identified, whereas for groups DN2 and DN3 no neurons have yet been identified in this volume.

Interestingly, almost all oviposition-related neurons have been identified in the hemibrain (only two oviDN on the left hemisphere are still missing), and very recently a couple of additional groups called U and G, comprising 2 and 5 neurons each, respectively, have also been shown to be important for oviposition and mapped to the connectome (Vijayan et al., 2023b). This makes a total of 25 oviposition-related neurons identified thus far.

To address the possibility of direct connections between clock neurons and neurons involved in egg laying, we probed the hemibrain connectome for these connections. The full results are displayed in Supplementary Tables 2 and 3, but in short, we found that whereas there are no connections between LNv or DN1 neurons and oviposition neurons, LNdS contact oviIN and pC1 neurons (Figure 5), further supporting the experimental findings described in previous sections. Interestingly, different subgroups of E cells (Shafer et al., 2022) show distinct patterns of connectivity to oviposition neurons: E3 neurons (comprising Cry-LNd1, LNd2 and LNd3) only contact the pC1 group, while E2 neurons (comprising Cry+ LNd4 and LNd5) only contact the oviIN neurons (Figure 5). On the other hand, the E1 neurons (comprising Cry+ LNd6 and 5-th sLNv) make no contacts with oviposition related neurons. This evidence suggests that the subset of E cells responsible for the loss of rhythmic oviposition are likely E2 neurons.

Interestingly, we noticed that connections between clock and oviposition neurons are bidirectional, although the E2-oviIN connection is stronger in the direction from the clock to oviposition neurons. This analysis suggests that, even though some feedback is anticipated, the E2 neurons are likely the ones providing temporal information to the oviposition circuit.

Additionally, aside from the circadian groups considered so far, we found strong, bidirectional connections between the clock neurons LPN (Reinhard et al., 2022) and the oviIN, with the strongest connections going from oviIN to LPN (Supplementary Tables 2 and 3 and Supplementary Figure S51).

Discussion

The oviposition rhythm is one of the less studied behavioral traits of *Drosophila*. One possible reason is merely technical, since egg collection and counting are a very laborious processes and there are no standard devices (just as the DAM system for locomotor activity) to reproducibly

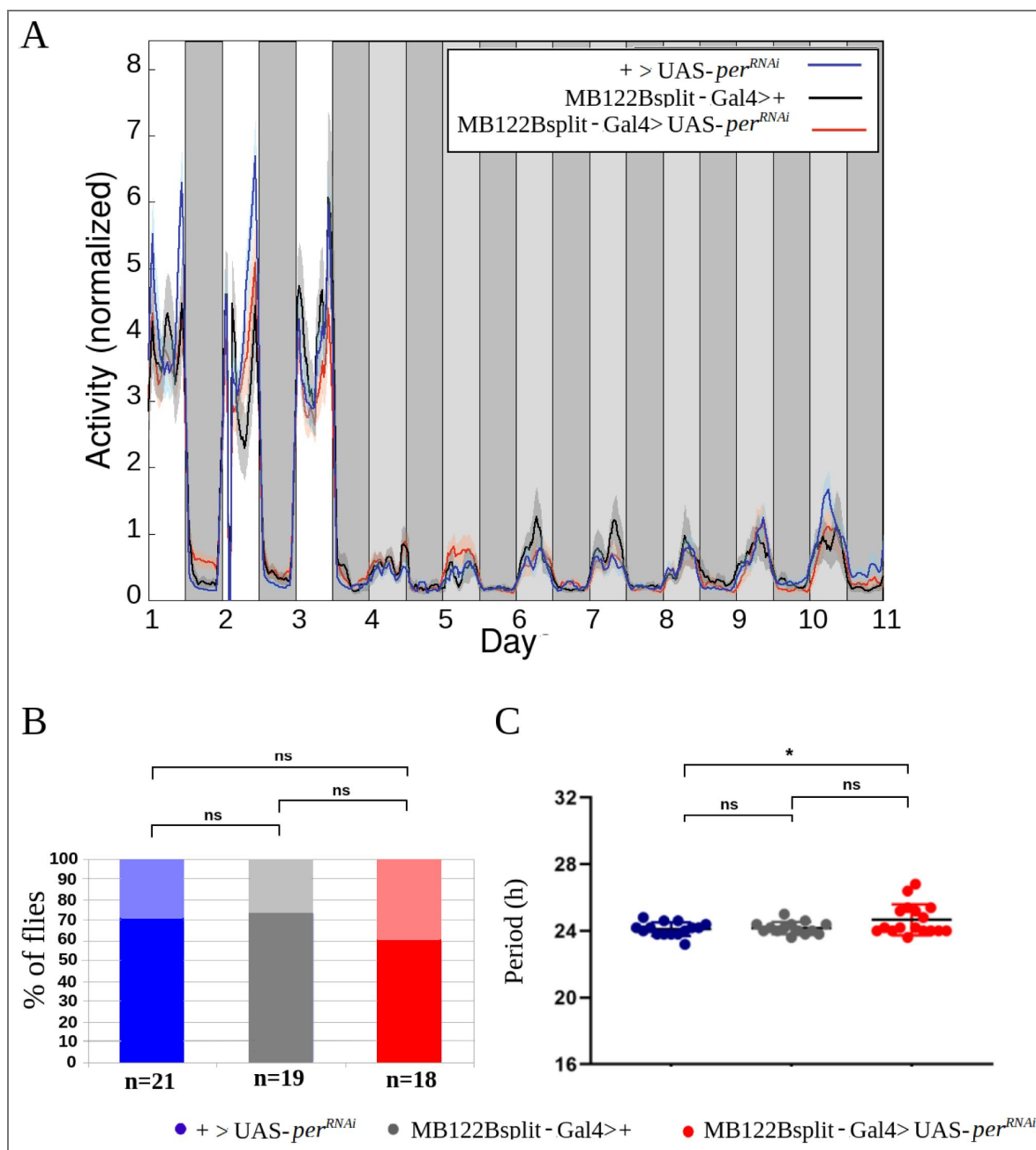


Figure 4. Disruption of the molecular clock in Cry+ LNd neurons does not alter the circadian rhythmicity of locomotor activity of mated female flies.

A: Average activity recording during 3 days in LD and 7 in DD. Light grey bars represent subjective days. **B:** Percent of rhythmic flies in DD. **C:** Periods of locomotor activity in DD of individually rhythmic flies. Mean $\pm$ SEM indicated. Each dot represents one fly. ns: non significant, *: $p < 0.05$ (chi-squared test for the comparison of proportions, and one-way-ANOVA with post-hoc Tukey for the scatter plots).

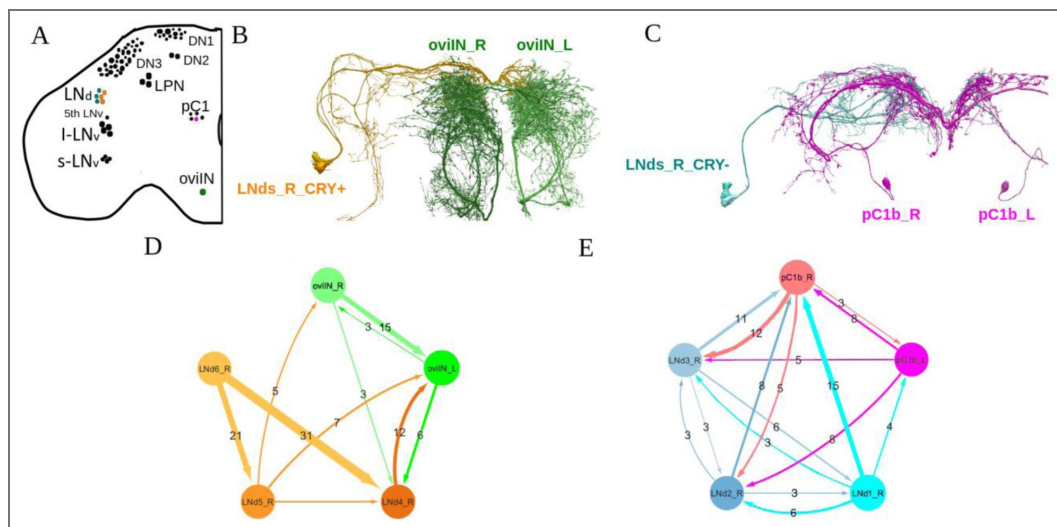


Figure 5. Direct synaptic connections between circadian clock neurons and oviposition-related neurons in the hemibrain dataset.

A: Schematic diagrams showing the different neuron clusters analyzed (in color). **B:** Connection between Cry+ LNd and ovilN neurons. **C:** Connection between Cry-LNd 2 and pC1b neurons. **D:** Network representation of the connectivity between Cry+ LNd and ovilN neurons. **E:** Network representation of the connectivity between Cry-LNd and pC1b neurons. Numbers give the number of synaptic contacts, which represent the strength of the connections. Only intermediate (between 3 and 9 synaptic contacts) and strong (>9 synaptic contacts) connections are considered.

evaluate it. Data analysis and rhythmicity assessments are also particularly difficult because the presence or absence of a rhythm must be established with only approximately 6 data points per cycle (in contrast to locomotor activity time series, where hundreds of data points per cycle are obtained), which can take only a few discrete values (typically, from 0 to 10). The analysis is further complicated by the fact that the daily average of eggs laid decreases with time (Kaufman and Demerec, 1942 [↗](#)). To tackle these limitations, we developed a semiautomated method for egg collection and a new pipeline for data analysis. We observed that the oviposition rhythm is less consolidated than locomotor activity rhythms. Surprisingly, this happens even in LD, where time of day cues are provided by light, the most important *zeitgeber* for diurnal animals. Egg-laying is even less rhythmic in DD since the information from the endogenous clock is one among many internal and external factors that influence the decision of laying an egg. In a sense, the oviposition record of a single fly can be considered as periodic signal with strong noise added to it. Such data might not even look rhythmic when analyzed individually. However, if we assume that the periodic component is the same for all flies in a population, and the noise is different for each individual, the periodic component can be extracted by averaging the oviposition data. This led us to prefer periodograms of averaged data for the assessment of the rhythmicity of a given genotype.

Even though there have been some advances in the understanding of the relationship between circadian clock and oviposition (Howlader and Sharma, 2006 [↗](#); Manjunatha et al., 2008 [↗](#)), there is still no information about the relative importance of the different neuronal groups in driving this behavior. Here, we downregulated the molecular clock in subsets of circadian neurons in order to establish their role in the control of circadian oviposition. First, we found that animals with downregulated *per* expression in all clock neurons of the brain display a drastic reduction of rhythmic oviposition, underscoring a key role of the central clock (as compared with peripheral clocks) in the control of the timing of oviposition.

Even though we also disrupted the molecular clock in PDF-expressing neurons as well as in the DN1p group, egg-laying rhythms were not abolished, in agreement with previous reports (Howlader et al., 2006a [↗](#)). We cannot rule out a contribution of PDF-expressing neurons, since we observed a shortening of the oviposition period. Interestingly, a previous work (Howlader et al., 2006b [↗](#)) reported a similar period shortening in *pdf⁰¹* mutant flies, but not when PDF-expressing neurons were ablated. Taken together, these results show that communication between either group and any other neurons is not necessary for rhythmic oviposition. This is in stark contrast with the situation for locomotor activity, which has been shown to become arrhythmic when PDF+ cells are ablated (Grima et al., 2004 [↗](#); Stoleru et al., 2004 [↗](#)). Interestingly, however, it has been recently shown that disrupting the molecular clock in these same neurons does not alter locomotor rhythmicity (Delventhal et al., 2019 [↗](#); Schlichting et al., 2019 [↗](#)) suggesting a more complex scenario.

Since it has been shown that the DN1p generate the rhythm of oogenesis (Zhang et al., 2021 [↗](#)), one might wonder why the disruption of their molecular clock or their electrical silencing does not also abolish the rhythm of oviposition. It is important to understand that the suppression of the oogenesis rhythm does not necessarily have an impact on rhythmic egg laying. The persistence of oviposition rhythm suggests that it does not depend on the rhythmic production of oocytes, but is instead an intrinsic property of the motor program which is the hypothesis more supported by our results.

Somewhat surprisingly, the disruption of the molecular clock in Cry + LNd neurons led to an almost complete loss of the oviposition rhythm. In contrast, downregulating the clock in Cry+ LNds did not affect rhythmic locomotor activity in mated females, underscoring a degree of specificity. Interestingly, in males the Cry+ LNds have been singled out as responsible for driving the activity peak at dusk (evening peak, (Guo et al., 2017 [↗](#))), which is the time when, in females, egg laying has a peak. As far as we know, oviposition is the first female-specific behaviour specifically controlled by the LNds. In males the LNds have been shown to be involved in the circadian control of the expression of sex-specific fat body genes (Erion et al., 2016 [↗](#); Fujii et al., 2008 [↗](#)).

The neural circuitry that controls the egg laying process has just begun to be unraveled (Vijayan et al., 2023a [↗](#); Wang et al., 2020 [↗](#); Zhou et al., 2014 [↗](#)). It includes the oviDNs, oviINs, oviENs, groups U and G, and pC1 neurons. Using the hemibrain connectome we found that neither the identified DN1s nor the LNV neurons directly contact any neurons ascribed to the oviposition circuit; the LND cluster, on the other hand, exhibits synaptic contacts with pC1 and oviIN neurons, providing the anatomical substrate for our experimental results. Furthermore, there is a clear separation of roles inside the LND cluster, since the oviINs are only contacted by two Cry⁺ LNDs (the so-called E2 cells (Shafer et al., 2022 [↗](#))) and the pC1s are only contacted by the three Cry⁺ LNDs (E3 cells (Shafer et al., 2022 [↗](#))). There is also a functional separation within the pC1 cluster, since the only pC1 neurons connected E3 neurons are the pC1b neurons from both right and left hemispheres (Figure 5 [↗](#)). The resulting clock/oviposition network is summarized in Figure 6 [↗](#).

Our data posits the E2 LNDs-oviIN connection as the candidate to mediate the timing of egg laying rhythms. The oviIN neurons are two of the most important neuronal hubs included in the current version of the *Drosophila* connectome, receiving strong inputs of hundreds of other neurons. Thus, the oviINs integrate the information from the circadian clock with the information coming from many other sources in order to signal to the oviposition circuit. Under such scenario the circadian clock would compete with many other sources to set the threshold for egg deposition (Vijayan et al., 2023a [↗](#)).

We hypothesize that the E3-pC1b connection is less important for conveying circadian information to the oviposition circuit because the pC1b are located upstream within the oviposition circuit that controls the motor program, being connected only to the oviIN neurons (Wang et al., 2020 [↗](#)). On the other hand, since pC1 neurons control female receptivity, it is tempting to speculate that this behavior may have a circadian rhythm, that could be generated through the E3-pC1b connection. Interestingly, the female mating rate has been shown to display circadian rhythmicity (Sakai and Ishida, 2001 [↗](#)), and very recently it has been proposed that pC1b neurons are involved in the control of female copulation rate (Li et al., 2023 [↗](#)). Conversely, given that the E3-pC1b connection is bidirectional, it could also be instrumental in conveying information about mating status to the circadian clock, therefore producing the changes in temporal organization that females undergo after mating (Delbare et al., 2023 [↗](#); Elwyn Isaac et al., 2010 [↗](#); Riva et al., 2022 [↗](#)).

The function of the LPN neurons in the circadian clock is still unclear (Reinhard et al., 2022 [↗](#)) and therefore no clear role in rhythmic oviposition can be ascribed to the LPN-oviIN connection. On the other hand, since there are many more oviIN inputs to the LPN than outputs (135 vs 16 synaptic connections), it seems more likely that this connection conveys information to the circadian clock from those processes in which the oviINs are involved.

The information provided by the *Drosophila* connectome is only the first step towards the understanding of the influence of the circadian clock on oviposition. Future work will be necessary to test the functionality of these connections as well as the role of different neuropeptides (sNPF, ITP, PDF) and neurotransmitters (such as acetylcholine and glutamate) in the control of egg laying behavior.

Material and methods

Fly strains

All fly strains used in this study are detailed in Table 1 [↗](#). Flies were reared and maintained on standard cornmeal/agar medium at 25 °C and 60% humidity in a 12 hr:12 hr LD cycle unless stated otherwise. For oviposition and locomotor activity experiments we used a different medium, named banana medium, prepared with 200g of banana, 25g of barley, 36g of black sugar, 35 g of yeast, 12.5g of agar and 2g of Nipagin per liter of water ¹⁵.

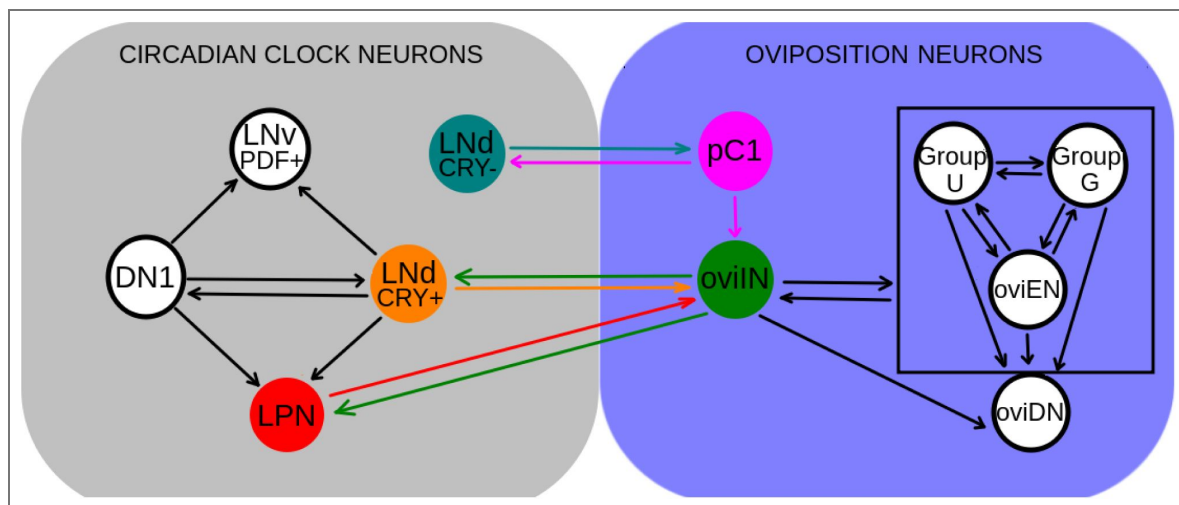


Figure 6. Scheme of connections between circadian clock and oviposition-related neuron clusters of the same hemisphere in the hemibrain dataset.

Each circle represents a neuron cluster, comprising different numbers of neurons (some clusters comprise only one neuron). Clusters and connections involved in the connectivity between circadian clock and oviposition sets have been colored. In each connection the arrow points to the post-synaptic cluster. OviIN neurons are bidirectionally connected to every neuron of the clusters inside the square.

<i>Drosophila melanogaster</i> line	Source	Identification number	Common denomination
Canton-S	Bloomington <i>Drosophila</i> Stock Center	BDSC: 64349	CS
Yellow white	Bloomington <i>Drosophila</i> Stock Center	BDSC: 1495	y w
y per ^S w	Donated by Jeff Hall	Collection of Jeff Hall	per ^S
y[1] sc[*] v[1] sev[21]; P{y[+t7.7] v[+t1.8]=TRiP.HMS 02045}attP2/TM3, Sb[1]	Bloomington <i>Drosophila</i> Stock Center	BDSC:40878	UAS-per ^{RNAi}
w[*];	Bloomington <i>Drosophila</i>	BDSC:93198	Clk856-Gal4

Table 1. All fly strains used in this study.

P{w[+mC]=Clk-GAL4.-856}2	Stock Center		
w[*]; sna[Sco]/CyO; P{w[+mC]=Clk-GAL4.1.5}4.1M/TM 6B, Tb[1]	Bloomington Drosophila Stock Center	BDSC:36316	Clk4.1-Gal4
P{w[+mC]=Pdf-GAL4.P2.4}X, y[1] w[*];Dicer/cyo	Bloomington Drosophila Stock Center	BDSC:6899	PdfDicer-Gal4
Mai179-Gal4;Pdf-Gal80	Donated by José Duhart	Collection of Patrick Emery	Mai179-Gal4;Pdf-Gal80
MB122B E-cell split-Gal4	Donated by José Duhart	Collection of Orie Shafer	MB122B-splitGal4
w;; kir2.1(1)/TM3 Sb	Donated by Justin Blau	Collection of Justin Blau	UAS-kir2.1
w;UAS-cyc ^{DN} ;+	Donated by Fernanda Ceriani	Collection of Sebastian Kadener	UAS-cyc ^{DN}

Table 1. (continued)

Semiautomated egg collection device

The egg collection device (Supplementary Figure 1 [↗](#)) consists of a wooden basis with a mechanical arm fixed to it, and plastic sets of tracks with fly chambers that can be replaced at will. Each “set of tracks” is a 13.9 x 19.5 cm 3D-printed piece with 7 tracks. Inside each track there are 6 equidistant wells measuring 15 x 20 mm, which are filled with banana medium. Each fly chamber (17 x 27 mm) has a transparent roof, and no floor, so that the fly inside it is in contact with the food wells, when the chamber is placed on the track so that it can slide along it. 3 sets of tracks are placed on the wooden base, and the chambers are moved together from well to well by a mechanical arm, which is moved by a stepper motor controlled by an Arduino UNO. The chambers are displaced from well to well every 4 hours. More frequent sampling gives rise to less consistent rhythmic patterns. Survival in this egg collection device is in general high: on average, more than 80% of the flies are alive at the end of an egg collection experiment lasting one week.

Behavioral assays

The egg deposition behavior of females was analyzed individually at 25°C and 60% humidity. Before starting the assay, six 0-5 day old virgin females and five males of the desired genotype were anesthetized using CO₂ and introduced into a vial with 10mL of standard food for 72 hours. This allowed for the crossing of 50-80 female flies with their respective males per experiment. Subsequently, the resulting gravid females from the crosses were placed individually in vials containing a plastic spoon with 1.5mL banana medium along with a drop of yeast for one day. The eggs deposited by each of the females are counted under a binocular stereoscopic magnifier (Lancet Instruments) and the 21 best egg layers are selected to conduct the circadian oviposition experiment. This prior selection of the best egg layers is performed because, of the total females put to mate with males, nearly half do not lay eggs or lay very few.

21 of these females were housed in the chambers of each apparatus, which was then introduced in an incubator at constant temperature (25 °C). The first 1 or 2 days the light regime was LD (12:12) in order for the flies to adapt to their new environment, and then they entered a DD regime for a given number of days. The sets of tracks were retired and replaced by sets with new food every 20, 24 or 28 hours (in order to avoid providing an entrainment signal to the flies). The eggs in each food well of each of the retired tracks were counted under a binocular stereoscopic magnifier (Lancet Instruments), and registered.

For the locomotor activity assays we used the setup used in previous works (Riva et al., 2022 [↗](#)). Briefly, the flies are housed in translucent tracks and their movement is registered with video cameras above the tracks, connected to a computer, where the position of each fly is extracted from the video (with ad hoc programmed software) and registered. Locomotor activity of mated females was monitored for 3 days in LD conditions and then transferred to DD conditions for 7 days. Every 4 or 5 days the flies were transferred to tracks with fresh food, because after that time the appearance of larvae hindered the video tracking of the females.

Data analysis

After each oviposition experiment lasting N days the data consisted of a time series E of $6N$ points for each fly. A new time series was generated by averaging the individual series. Since the number of eggs laid by a mated female tend to show a downward trend, we proceeded as follows, in order to detrend the data (see the Supplementary Material for further details). First, a moving average of the data is performed, with a 6 point window, and a new time series T is obtained. In principle, T is a good approximation to the trend of the data. Then, a new, detrended, time series D is generated by pointwise dividing the two series (i.e. $D(i)=E(i)/T(i)$, where i indexes the points of each series).

In order to assess the rhythmicity of a genotype, we averaged the detrended time series and performed a Lomb-Scargle periodogram (Ruf, 1999 [↗](#); VanderPlas, 2018 [↗](#)). For this we used package Lomb (version 2.1.0), from the R Statistical Software (v4.1.2; R Core Team 2021). The

significance lines were calculated by repeatedly randomizing the time series (using function *randslp* in Lomb R package). A genotype was considered rhythmic if the periodogram had one peak between 16 and 32 hs, and it was above the $p=0.05$ significance line.

The graphs showing individual periods were made by obtaining the individual periods using the Lomb-Scargle periodogram described above for each particular individual. For the individual periods shown in the figures, we added the weakly rhythmic category, corresponding to flies whose periodograms displayed only one peak between 18 and 32 hs, and a power larger than 0.2.

More details about the method used can be found in the Supplementary Material 2.

For locomotor activity experiments, these files obtained from the video tracking were processed with an analysis software we developed (in Bash) which provides statistics for activity (position, distance traveled, etc.) (Riva et al., 2022 [DOI](#)).

Analysis of connectome data

The data used to determine the connectivity between the circadian clock and the oviposition circuits come from the Hemibrain dataset (version 1.2.1) made publicly available by Janelia Research Campus (Scheffer et al., 2020 [DOI](#)). To access the data we used the NeuPrintExplorer Web tool (<https://neuprint.janelia.org/> [DOI](#)). The result of our queries is summarized in Supplementary Tables 2 and 3 [DOI](#). The neuron IDs are as given in the hemibrain dataset, just as the instance names of the oviposition related neurons. For circadian clock neurons we used the names given by Shafer et al. (Shafer et al., 2022 [DOI](#)). We only replaced LPN-4 by LPN-L to stress the fact that this neuron is placed in the left hemisphere (all the others are in the right hemisphere). The strength of each connection was quantified using the criteria of Scheffer et al. (Scheffer et al., 2020 [DOI](#)). Connections with more than 9 synapses were considered strong, connections having between 3 and 9 synapses were considered as having intermediate strength, and connections with 0,1, or 2 synapses were considered as weak. Because weak connections are prone to error (Scheffer et al., 2020 [DOI](#)) we excluded them from the analyses. The figures that show connectivity between sets of neurons were drawn using Cytoscape version 3.10.1 (Shannon et al., 2003 [DOI](#)).

Statistical analysis

The following statistical analyses were used for the comparison between genotypes of individual periods (scatter plots). First, we tested data for normality with D'Agostino & Pearson test. Once the normality of the data was confirmed, we used a unpaired t-test for test differences between means for experiments where 2 groups were compared and a one-way ANOVA for experiments where 3 groups were compared. When data was not normally distributed, we applied a Mann-Whitney test for test differences between means. When significant differences were found, we conducted a post-hoc Tukey test with a correction for multiple comparisons. In all the analyses we used Two-Tailed p -values. To perform all the tests, we used GraphPad Prism (Boston, Massachusetts USA, www.graphpad.com [DOI](#)) and R Statistical Software (v4.1.2; R Core Team 2021). $p < 0.05$ was considered statistically significant. In dot plots, horizontal lines represent the mean value; error bars depict the standard error of the mean. For the comparison between genotypes of the proportion of rhythmic individuals over the total (proportion plots), we used the 'N-1' Chi-squared test as recommended by Campbell, 2007 [DOI](#) and Richardson, 2011 [DOI](#). To perform these tests, we use MedCalc Software Ltd. Comparison of proportions calculator (https://www.medcalc.org/calculator/comparison_of_proportions.php [DOI](#)) (Version 23.2.7; accessed June 19, 2025). No statistical methods were used to determine sample size. Sample sizes are similar to those generally used in this field of research. Samples were not randomized and analyzers were not blind to the experimental conditions. In all the figures we show results of two or three independent experiments.

Data availability

All data is available upon request.

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Additional information

Author contributions

Conceptualization, Methodology and Validation: S.R; S.R.G; D.L.F; Formal Analysis, Investigation, and Visualization, S.R; S.R.G; D.L.F; Writing – Original Draft, S.R; S.R.G; D.L.F; Writing – Review & Editing: S.R, M.F.C; S.R.G, D.L.F ; Supervision, S.R.G, D.L.F ; Project Administration and Funding Acquisition, M.F.C, S.R.G., D.L.F.

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Additional files

[Supplementary file 1](#) 

[Supplementary file 2](#) 

[Supplementary file 3](#) 

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Peer reviews

Joint Public review:

Summary

Riva et al. introduce a semi-automatic setup for measuring *Drosophila melanogaster* oviposition rhythms and use it to map the timekeeping function underlying egg laying rhythms to a subset of clock cells. Using a combination of neurogenetic manipulations and referencing the publicly available female hemi-brain connectome dataset, they narrow the critical circuit down to possibly two of the three CRYPTOCHROME expressing lateral-dorsal neurons (LNs). Their findings suggest that different overlapping sets of clock neurons may control different behavioral rhythms in *D. melanogaster*.

This work will be of interest to researchers interested in the circadian regulation of oviposition in *D. melanogaster* (and possibly other insects), a phenomenon which has been left relatively under-explored. The construction of a semi-automated setup which can be made relatively cheaply using available motors and 3D printed molds provides a useful model for obtaining longer records of oviposition activity. The analysis of noisy oviposition timeseries, however, may require revisiting both the methods used for sampling eggs laid per female as well as the analytical tools used to clean up and analyze individual records, because simple averaging can lead to incorrect conclusions regarding the underlying nature of the rhythm.

Strengths

Additional experiments were carried out for this revised version of the manuscript that strengthen their original findings. These include: using a dominant negative form of the circadian clock gene, *cycle*, to disrupt the circadian clock, which provides additional support for the role of CRY+ LNs in generating the circadian rhythm of oviposition; reassessing the functionality of PDF neurons and showing that they seem to be important for maintaining the circadian period of egg laying; using the *per01* mutation to show the role of period locus function in the control of the circadian rhythm of oviposition. The authors also point to some potentially interesting connectome data that suggest hypotheses regarding the neuronal circuit linking daily timekeeping to oviposition, which will require further validation in future studies. The videos and pictures demonstrate the working of the semi-automated egg collection setup, which should help others create similar devices.

Weaknesses:

The major weaknesses of this work result from the noisy nature of the data.

They include:

(1) Problems associated with averaging: The authors intended to focus on the oviposition clock in individual females, however due to the inherent noise in the oviposition rhythm they had to resort to averaging across Lomb-Scargle periodograms generated from individual time-series. They then tested whether the averaged periodogram contains a significant frequency. However, this reduction in noise also reduces the ability to compare differences in power of the rhythm across individuals. Furthermore, this method makes it especially difficult to distinguish the contribution of subsets of the circuit on the proportion of rhythmic flies and the power of the rhythm. In this revised version the authors use two manipulations to disrupt the molecular clock, which could have different success rates based on the type and number of cells targeted. Unfortunately, the type of averaging used prevents the detection of any such effects. It is to be noted that, indeed, individual-level differences in period between the PdfDicer-Gal4 > perRNAi and UAS-perRNAi lines help the authors to

establish that there is a significant reduction in period length when the molecular clock is abolished in PDF cells. These individual measurements are now very helpful in discerning the effect of manipulations carried out on different circadian neural subsets, some of which could have been missed if only averages were considered.

(2) Sensitivity to sample size: Averaging reduces the effect of random background noise but noise reduction is dependent upon sample size. Comparing genotypes with different sample sizes in addition to varying signal to noise ratios (which might also change with neural manipulations) makes it difficult to estimate how much of the rhythm structure is contributed by a given neuronal subset; thus, whenever possible comparisons should be made between groups that include similar number of flies. This problem is compounded when the averaged periodogram is composed of both rhythmic and weakly rhythmic individuals. For instance, in the main text the reported value of period length of pdfDicer-Gal4 > perRNAi is 20.74h (see also Fig 2J) but in the Supplementary figure 2S1 this is close to 22h, while the values reported for the control are largely similar (24.35h in Fig 2H versus ~24h in Fig 2S1). A difference of 3.6h between control and experimental flies is much greater than 2h. Which estimate (average versus individual) is more reliable in predicting the behavior of these flies is difficult to determine without further experiments.

(3) Based on the newly provided data for individual fly periodograms the reader can visually evaluate the rhythmicity associated with each genotype. Such visual inspection did not reveal any clear difference between the proportion of rhythmic individuals between experimental and parental GAL4 and/or UAS controls, except for experiments using per01 mutant animals. This is surprising since if these circuits are controlling the oviposition rhythm, perturbing them should affect most individuals in a similar way.

In summary, although the authors have implicated CRY+ LNdS in the generation of a circadian rhythm in oviposition it is not clear looking at individual readouts if this manipulation is rendering flies arrhythmic or changing the period of the clock slightly, such that there is increased variation in period length at the individual level which is not being captured by the low signal to noise ratio and in the average gives a flattened output as a result. Thus, while the manipulations done to the clock in these neurons might indeed affect the circadian nature of the oviposition rhythm it is still rather difficult to determine if they are indeed the sole clock cells generating this rhythm especially when nearby PDF+ cells also affect period length. Nevertheless, the connectomic data do show that they are very close to the OviIN neurons, placing them at an important juncture of transmitting circadian time information to the downstream oviposition circuit. Overall, the authors have achieved some of their aims, although the analysis methods leave some of their inferences open to speculation.

Other comments

Disrupting the clock in the 5th sLNv and 3 Cry+ LNdS (and weakly in a small subset of DN1) affected egg-laying. Although the work emphasizes the importance of the LNd, the role of the 5th sLNv's role should be discussed.

<https://doi.org/10.7554/eLife.103359.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Joint Public review:

Weaknesses:

(1) Controls for the genetic background are incomplete, leaving open the possibility that the observed oviposition timing defects may be due to targeted knockdown of the period (per) gene but from the GAL4, Gal80, and UAS transgenes themselves. To resolve this issue the authors should determine the egg-laying rhythms of the relevant controls (GAL4/+, UAS-RNAi/+, etc); this only needs to be done for those genotypes that produced an arrhythmic egg-laying rhythm.

(2) Reliance on a single genetic tool to generate targeted disruption of clock function leaves the study vulnerable to associated false positive and false negative effects: a) The per RNAi transgene used may only cause partial knockdown of gene function, as suggested by the persistent rhythmicity observed when per RNAi was targeted to all clock neurons. This could indicate that the results in Fig 2C-H underestimate the phenotypes of targeted disruption of clock function. b) Use of a single per RNAi transgene makes it difficult to rule out that off-target effects contributed significantly to the observed phenotypes. We suggest that the authors repeat the critical experiments using a separate UAS-RNAi line (for period or for a different clock gene), or, better yet, use the dominant negative UAS-cycle transgene produced by the Hardin lab (<https://doi.org/10.1038/22566>).

We have followed the referee advice, repeating the experiments with the dominant negative UAS-cyc^{DN}. They nicely confirm our conclusions: the abolition of the cellular clock in LNd neurons rule out the rhythmicity of oviposition. The results are presented in Fig. 3 of the new manuscript, panels H to N. We thank the reviewer for this suggestion that has definitely improved our paper, since it allows us to confirm our result using both a different driver and a different UAS sequence. In addition, we included the required GAL4 controls, which can be found in Panels E, L of the figure as well as average egg laying profiles for all genotypes involved (Panels B, D, F, I, K and M). Regarding the MB122Bsplit-Gal4>UAS-per^{RNAi} experiment, we moved it to a supplementary figure (Figure 3S1). The paragraph where the new Figure 3 is discussed has been modified accordingly.

(3) The egg-laying profiles obtained show clear damping/decaying trends which necessitates careful trend removal from the data to make any sense of the rhythm. Further, the detrending approach used by the authors is not tested for artifacts introduced by the 24h moving average used.

The method used for the assessment of rhythmicity is now more fully explained and tested in the supplementary material. In particular, the issue of trend removal is treated in the second section of the SM, and the absence of "artifacts" (interpreted as the possibility of deciding that a signal is rhythmic when it is not, or vice versa) shown in figs. S3 to S5.

(4) According to the authors the oviposition device cannot sample at a resolution finer than 4 hours, which will compel any experimenter to record egg laying for longer durations to have a suitably long time series which could be useful for circadian analyses.

The choice of sampling every 4 hours is not due to a limitation imposed by the device used. In fact the device can be programmed to move at whatever times are desired. As mentioned in the Material and Methods section, "more frequent sampling gives rise to less consistent rhythmic patterns", because the number of eggs sampled at each time slot become too small. In particular, we have tested sampling at intervals of 2 hours, and we have observed that this doubles the work performed by the experimenter but does not lead to an improvement in the assessment of rhythmicity.

(5) Despite reducing the interference caused by manually measuring egg-laying, the rhythm does not improve the signal quality such that enough individual rhythmic flies could be included in the analysis methods used. The authors devise a workaround by

combining both strongly and weakly rhythmic ($LSpower > 0.2$ but less than $LSpower$ at $p < 0.05$) data series into an averaged time series, which is then tested for the presence of a 16-32h "circadian" rhythm. This approach loses valuable information about the phase and period present in the individual mated females, and instead assumes that all flies have a similar period and phase in their "signal" component while the distribution of the "noise" component varies amongst them. This assumption has not yet been tested rigorously and the evidence suggests a lot more variability in the inter-fly period for the egg-laying rhythm.

As stressed in the paper, and in the new Supplementary Material, the individual egg records are very noisy, which in general precludes the extraction of any information about the underlying period and phase. The workaround we (and others, e.g. Howlader et al. 2006) have used is analyzing average egg records for each genotype. Even though this implies assuming the same period and phase for all individuals, we have observed, using experiments with synthetic data, that small variations in individual periods (of the same amount as those present in real experiments where the period of some flies can be assessed individually) still allow us to use our method to decide if the genotype is rhythmic or not. This issue is discussed at length in the new Supplementary Material. There we also discuss an experiment with real flies, showing the individual records, and the corresponding periodograms, for each fly, for a rhythmic (Fig. S14) and an arrhythmic genotype (Fig. S17).

*(6) This variability could also depend on the genotype being tested, as the authors themselves observe between their Canton-S and YW wild-type controls for which their egg-laying profiles show clearly different dynamics. Interestingly, the averaged records for these genotypes are not distinguishable but are reflected in the different proportions of rhythmic flies observed. Unfortunately, the authors also do not provide further data on these averaged profiles, as they did for the wild-type controls in Figure 1, when they discuss their clock circuit manipulations using *perRNAi*. These profiles could have been included in Supplementary figures, where they would have helped the reader decide for themselves what might have been the reason for the loss of power in the *LS* periodogram for some of these experimental lines.*

We have added the individual periodograms of the arrhythmic lines to the Supplementary material (Figs. 3S2, 3S5 and panel G of Fig. 3S1), where they can be compared with their respective controls (Figs 3S3, 3S4, 3S6, 3S7 and panel F of Fig. 3S1).

(7) By selecting 'the best egg layers' for inclusion in the oviposition analyses an inadvertent bias may be introduced and the results of the assays may not be representative of the whole population.

We agree that the results may be biased for 'the best egg layers'. We remark however, that the flies that have been left out lay very few eggs, some of them even laying no eggs on a whole day. For these flies it is difficult to understand how one can even speak of egg laying rhythmicity (let alone how one can experimentally assess it). Thus, we think it might be misleading to speak of results as "representative of the whole population". Furthermore, it is even possible that the very concept of egg laying rhythmicity makes little sense if flies do not lay enough eggs.

(8) An approach that measures rhythmicity for groups of individual records rather than separate individual records is vulnerable to outliers in the data, such as the inclusion of a single anomalous individual record. Additionally, the number of individual records that are included in a group may become a somewhat arbitrary determinant for the observed level of rhythmicity. Therefore, the experimental data used to map the clock neurons responsible for oviposition rhythms would be more convincing if presented alongside individual fly statistics, in the same format as used for Figure 1.

In general, we have checked that there are no "outliers", in the sense of flies that lay many more eggs than the others in the experiment. But maybe the reviewer is referring to the possibility that a few rhythmic flies make the average rhythmic. This issue is addressed in the supplementary material, at the end of section "Example of rhythmicity assessment for a synthetic experiment". In short, we found that eliminating some of the most rhythmic flies from a rhythmic population makes the average a bit less rhythmic, but still significantly so. Conversely, if these flies are transferred to an arrhythmic population, the average is still non rhythmic.

Regarding "the number of individual records that are included in a group may become a somewhat arbitrary determinant for the observed level of rhythmicity", we stress that we have not performed a selection of flies for the averages. All of the flies tested are included in the average, independently of their individual rhythmicity, provided only that they lay enough eggs.

(9) The features in the experimental periodogram data in Figures 3B and D are consistent with weakened complex rhythmicity rather than arrhythmicity. The inclusion of more individual records in the groups might have provided the added statistical power to demonstrate this. Graphs similar to those in 1G and 1I, might have better illustrated qualitative and quantitative aspects of the oviposition rhythms upon per knockdown via MB122B and Mai179; Pdf-Gal80.

We are aware that in the studies of the rhythmicity of locomotor activity the presence of two significant peaks is usually interpreted as a "complex rhythm", i.e. as evidence of the existence of two different mechanisms producing two different rhythms in the same individual. In our case, since the periodograms we show assess the rhythmicity of the average time series of several individuals, the two non-significant peaks could also correspond to the periods of two different subpopulations of individuals. However, a close examination of the individual periodograms, now provided as Supplementary Figures 3S2 to 3S9, does not show any convincing evidence of any of these two possibilities.

Another possibility could be that such peaks are simply an artifact of the method in the analysis of time series that consist of very few cycles and also few points per cycle. In the supplementary material we show that this can indeed happen. Consider, for example, periodograms 2 and 4 in Fig. S12 of the SM. Even though both of them display two non significant peaks, these periodograms correspond to two synthetic time series that are completely arrhythmic.

We have added to the manuscript a paragraph discussing the issue of possible bimodality (next to last paragraph in subsection "The molecular clock in Cry+ LNd neurons is necessary for rhythmic egg-laying").

Wider context:

*The study of the neural basis of oviposition rhythms in *Drosophila melanogaster* can serve as a model for the analogous mechanisms in other animals. In particular, research in this area can have wider implications for the management of insects with societal impact such as pests, disease vectors, and pollinators. One key aspect of *D. melanogaster* oviposition that is not addressed here is its strong social modulation (see Bailly et al., *Curr Biol* 33:2865-2877.e4. doi:10.1016/j.cub.2023.05.074). It is plausible that most natural oviposition events do not involve isolated individuals, but rather groups of flies. As oviposition is encouraged by aggregation pheromones (e.g., Dumenil et al., *J Chem Ecol* 2016 <https://link.springer.com/article/10.1007/s10886-016-0681-3>) its propensity changes upon the pre-conditioning of the oviposition substrates, which is a complication in assays of oviposition rhythms that periodically move the flies to fresh substrate.*

We agree that social modulation can be important for oviposition, as has been shown in the paper cited by the reviewer. But we think that, in order to understand the contribution of social modulation to oviposition, it is important to know, as a reference for comparisons, what the flies do when they are isolated. Our aim in this work has been to provide such a reference.

Recommendations for the authors:

(1) *The weaknesses identified in the Public review could be addressed as follows: etc.*

We have followed the suggestions of the editor and addressed each of the weaknesses mentioned (see details above).

(2) *Could the authors comment on their choice of using individual flies for their assay rather than (small) groups of flies? Is it possible that their assay would produce less noisy results with the latter?*

First we want to emphasize that our aim here was to assess the presence of individual rhythmicity, free from any external influences, whether arising from environmental external cues (such as light or temperature changes) or by social interactions (with other females or males). However, we were also curious about the behavior when males were put in the same chamber with each female. We performed a few tests and the results were very similar to what we obtained with single females.

(3) *Minor points:*

(a) *Line 57-58 - "around 24 h and a peak near night onset (Manjunatha et al., 2008). Egg-laying rhythmicity is temperature-compensated and remains invariant despite the nutritional state": Rephrase to something simpler like temperature and nutrition compensated.*

Corrected.

(b) *Line 56-57 - "The circadian nature of this behavior was revealed by its persistence under DD with a period around 24 h and a peak near night onset (Manjunatha et al., 2008)." A better reference here would be to Sheeba et al, 2001 for preliminary investigations into the egg-laying rhythms of individual flies and McCabe and Birley, 1998 for groups of flies under LD12:12 and DD.*

Suggestion accepted.

(c) *Line 65-67 - "We determined..... molecular clock in the entire clock network reduced the LNV did not." This suggests that it was unknown until now that LNV does not have a role, whereas Howlader et al 2006 already suggested that. The reader becomes aware of this at a later part of the manuscript. Please revise.*

This has been revised, and the citation to Howlader et al 2006 added to the new sentence.

(d) *Line 67 - "impairing the molecular clock in the entire clock network reduced the circadian rhythm of.."; saying "Reduced the power of the circadian rhythm" might be better phrasing."*

Suggestion accepted.

(e) *Line 72 - using the Janelia hemibrain dataset.*

Corrected

(f) Line 72 typo "ussing", should be 'using'.

Corrected.

(g) Line 94: why is the periodic signal the same for all on the first day of DD?

It is well known that in LD conditions activity is driven by the environmental light-dark cycle, which entrains the endogenous circadian clock of all flies. Even after the transition to DD, the effects of this entrainment persist for a few days, allowing the individual rhythmic patterns set by the light-dark cycle to remain synchronized for at least a few cycles. We are assuming that the same happens with oviposition. A sentence has been added explaining this (beginning of third paragraph of subsection "Egg-laying is rhythmic when registered with a semiautomated egg collection device").

(h) Figure 1A-D, Were all flies included or only rhythmic flies? Please make this clear. How do you distinguish rhythmic and arrhythmic flies in Figure 1E? Their representative individual plots of egg number graphs are required. Why was the number of flies under DD decreased from 20 to 18?

Throughout the paper, the analysis of average rhythmicity has been performed including all flies, since we postulate that even flies that individually can be classified as non rhythmic have a rhythm that is corrupted by noise, and that this noise can be partially subtracted by performing an average. The explanation of the characterization of rhythmic and arrhythmic individuals is in the Methods section, under the Data Analysis subsection. This is now fully developed in the Supplementary material, where the individual plots for some of the genotypes are included.

Regarding the question of the number of flies having "decreased from 20 to 18?", there is a misunderstanding here. The results depicted in Figure 1, and in particular in panel E, correspond to two different experiments: one performed only in LD (7 days, n=20), and a second one performed for 5 days in DD, with one previous day in LD (n=18).

(i) Figure E and K, Are n=20, 18, and n=30, 22 the total numbers of flies including both rhythmic and nonrhythmic? If so, it would be better to put them in the column, not in the rhythmic column.

The figure has been corrected.

(j) Line 107-108, please provide a citation for this statement.

We have added two references: Shindey et al. 2016, and Deppisch et al. 2022.

(k) Figure 1, 2, etc., please write a peak value inside the periodogram graph. This makes comparison easier.

The peak values have been added in all Figures.

(l) Line 184-185, Figure 2F, tau appears shorter in *Clk4.1>perRNAi* flies than in control, which suggests that *DNp1* may play a role?

As explained in the Supplementary Material, the particularities of oviposition records (discrete values, noise, few samples per period, etc.) preclude an accurate determination of the period if the record is considered as rhythmic. In particular, Fig. S4 shows that differences of 1 hour between the real and the estimated periods are not unusual.

(m) Figure 4. Why are 2 controls shown? Please explain. Are they the same strains?

The two controls shown are the UAS control and the GAL4 control. This information has now been added to the figure.

| (n) Line 314 'that' should be 'than'?

Corrected.

| (o) Line 73-74 - Phrasing is not clear in: "LNds and oviposition neurons, consisting with, the essential role of LNds neurons in the control of this behavior.""

Corrected.

| (p) Line 81-84 - "the experiments particularly demanding and labor-intensive. In this approach, eggs are typically collected every 4 hours (sometimes also every 2 hours), which usually implies transferring the fly to a new vial or extracting the food with the eggs and replacing it with fresh food in the same vial (McCabe and Birley, 1998; Menon et al., 2014)." McCabe and Birley had an automated egg collection device designed for groups of flies, which sampled eggs laid every hour for 6 days. Please remove this reference in this context

Reference removed.

| (q) Line 91-92 - "The assessment of oviposition rhythmicity is challenging because the decision of laying an egg relies on many different internal and external factors making this behavior very noisy." This sentence makes it appear that 'assessment' is the limitation. Even locomotor activity is governed by many internal and external factors, yet we can obtain very robust rhythms. The sentence that follows is also not easy to digest. Can the authors frame the idea better?

We have rewritten the corresponding paragraph in order to make it more clear (second paragraph of the Results section). Additionally, the Supplementary Material contains now a more detailed explanation and analysis of the method used.

| (r) Line 104-107 - rhythmic (with a period close to 24 h, Figure 1F) although the average egg record is strongly rhythmic with a period around 24 h (Figure 1B). Under DD condition, individual rhythmicity percentages are the same as in LD (Figure 1E) and their average record is also very rhythmic with a period of 24 h (Figure 1D). 'Strongly rhythmic' and 'very rhythmic' are less indicative of what is happening with the oviposition rhythm and can be phrased as robust instead, with a focus on their power measured.

We have accepted the suggestion.

| (s) Line 108-110 - "Thus, egg-laying displays a much larger variability than locomotor activity, compounding the difficulty of observing the influence of the circadian clock on this behavior." The section discussed here does not illustrate the variability in egg-laying as much as the lack of robustness of the rhythm. The variation in rhythmicity going from CS flies (~70% rhythmic) to yw flies (~50% rhythmic) showcases the variability in this rhythm and how it is difficult to observe when compared to locomotor rhythms, which are usually consistently >90% rhythmic across multiple genotypes. These lines can be placed after the discussion about yw and perS flies. Moreover, previous studies using individual flies have reported that egg-laying rhythm is more variable than others Figure 1, Sheeba et al 2001.

We have accepted the suggestion, replacing "Thus, egg-laying displays a much larger variability than locomotor activity..." by "This shows that, at the individual level, egg-laying is much less robust than locomotor activity ..."

(t) Figure 1. Genotype notation within the figure panels is not consistent with the accepted / conventional notation or with the main text or legend notations throughout the manuscript.

We are sorry for this mistake. We have corrected the genotype names in Figures and text in order to make notation consistent across the paper.

(u) Supplementary Figure 1 Legend. Error in upper right corner? Not left corner? The photo does not clearly show the apparatus. The authors may wish to consider clearer images and more details about the apparatus including details of the 3D printing of the device and perhaps even include a short video where the motor moves the flies to a new chamber (This is only a suggestion to advertise the apparatus, not related to the review of the manuscript). They could also provide information about what fraction of females survived till the end of each trial when 21 flies were examined with 4-hour sampling across 4-5 cycles.

In general, more than 80% of the females are alive at the end of a one week oviposition experiment. We have added this information in the Methods section at the end of the corresponding subsection ("Automated egg collection device"). Regarding the eggcollection device, we have replaced the photographs in what is now Supplementary Figure 1S1, and a short supplementary movie showing its operation.

(v) The results depicted in Figure 2B are that of averaged time series. Hence the reader does not know 'the fact' that knocked-down animals are not completely rhythmic. Is the "not completely arrhythmic" in reference to flies with a power > 0.2 (weakly rhythmic) in their egg-laying rhythm or to the presence of ~40% of male flies (Supplementary Table 1) with a locomotor rhythm after *perRNAi* silencing of most of their clock neurons? This is confusing because no intermediate category of flies is discussed in Figure 2. Please edit for clarity.

We were referring to the rhythmicity of the genotype, not of the individuals. We have rewritten the corresponding paragraph in order to make it clearer (last paragraph of the first subsection of the Results section).

(w) Line 173 - ablation or electrically silencing all PDF+ neurons (Howlader et al., 2006). There were no experiments carried out using electrical silencing of PDF+ neurons in the referenced paper.

We are sorry for this mistake. This has been corrected (we have deleted the mention to electrical silencing).

(x) Line 173 - Shortening of period by nearly 3 hours cannot be considered minor.

We agree, and we have deleted the word "minor".

(y) Line 332-333 - "We also disrupted the molecular clock (or electrically silenced) in PDFexpressing neurons as well as in the DN1p group with no apparent effect on egg-laying rhythms". There was period shortening observed for *pdf GAL4 > perRNAi* manipulation so there was an effect on the egg-laying rhythm. Additionally, *perRNAi* based silencing does not electrically silence PDF neurons as the *kir 2.1* was expressed only using *Clk4.1 GAL4* in the *Dn1ps*. This line should be rewritten.

We have rewritten the paragraph mentioned (third paragraph of the Discussion) in order to make it more accurate.

(4) Page 22 - Data Analysis

Since the number of eggs laid by a mated female tend to show a downward trend, we proceeded as follows, in order to detrend the data (see the Supplementary Material for further details). First, a moving average of the data is performed, with a 6 point window, and a new time series T is obtained. In principle, T is a good approximation to the trend of the data. Then, a new, detrended, time series D is generated by pointwise dividing the two series (i.e. $D(i)=E(i)/T(i)$, where i indexes the points of each series)." Can the authors provide a reference for this method of detrending? Smoothing can frequently introduce artifacts in the data and give incorrect period estimates. Additionally, the trend visible in the data, especially in Figure 1, suggests a linear decay that can be easily subtracted. Also, there is no discussion of detrending in the Supplementary material attached.

We are sorry for the confusion with the Supplementary materials. The method used for subtracting both noise and trend from the data is now fully explained in the new Supplementary Material. All the issues raised by the reviewer in this comment have been addressed there.

(5) Figure by figure

Page - Type (Figure or text) - Comment

(a) Page 6 Figure 1C There is remarkable phase coherence seen in the average egg laying time series for CS flies 5 days into DD and as the authors note in Lines 94-95 in the text "Under light-dark (LD) conditions, or in the first days of DD, it can be that the periodic signal is the same for all flies". Since this observation is crucial to constructing the figures seen later in the paper, a note should be made about why this rhythm could persist across flies, so deep into DD.

As mentioned above, we have added a couple of lines explaining why we think that the assumption of a synchronized periodic signal is reasonable, at least during the first cycles (second paragraph of the first subsection of section Results).

(b) Figure 1 G The effect of period/phase decoherence seems to be showing up here in the average profile for yw flies as they seem to completely dampen out after 2 days in DD and yet have a 24-hour rhythm in the averaged periodogram. The authors should make a note here if the LS periodogram is over-representing the periodicity of the first few days in DD or if comparing the first 3 vs. the last 3 days in DD gives different results.

The dampening observed in average oviposition records is a product of the dampening of the oviposition records, which is well known phenomenon, probably caused by the depletion of sperm in the female spermatheque. One of the aims of the method used in the paper was to avoid the bias introduced by this dampening, by means of a detrending procedure. This is explained in the Materials and Methods, and now full details are given in the new Supplementary Materials.

(c) Figure 1E, K Is this data pooled across 2-3 experiments, as discussed in lines 500-01 under 'Statistical Analysis'? Also, what test is being performed to check for differences between proportions here, seeing as there are no error bars to denote error around a mean value and no other viable tests mentioned in Statistical Analysis?

We are sorry for this omission. For the comparison of proportions we used the 'N-1' Chisquared test. We have added a sentence detailing this at the end of the Statistical analysis section.

(d) Figure 1 F, L Can the total number of weakly and strongly rhythmic values be indicated in the scatter plot?

Corrected.

(e) Figure 1F, L (legend) Is the Chi-squared test being performed on the proportion values of Figure 1(E, K) or for Figure 1(F, L)?"

The chi-squared test mentioned was used for Fig1 F-L. As explained above, for the comparison of proportions we used 'N-1' Chi-squared test. This has now been added to the legend of the figure

(f) Page 8 Figure 2B Seeing as individual flies with a LS periodogram power < 0.2 are considered weakly rhythmic in Figure 1 F, L can Clk856 > perRNAi flies on average also be considered weakly rhythmic, as the peak in the periodogram is above 0.3?

We prefer to use the weakly rhythmic class only for individual flies. Nevertheless, we agree that this periodogram shows that the genotype analyzed is not completely arrhythmic, and that this might be due to some remaining individual rhythmicity. As mentioned above, we have rewritten the last paragraph of the first subsection of section Results in order to discuss this.

(g) Figure 2D Can the authors comment on why there is a shorter period rhythm when PDF neurons have a dysfunctional clock, whereas previous evidence (Howlader et al., 2004) suggested that these neurons play no role in egg-laying rhythm? They should also refer to McCabe and Birley, 1998 to see if their results (where they observed a shorter period of ~19h with groups of per0 flies), might be of interest in their interpretations.

We have added a line commenting this in the corresponding subsection ("LNv and DN1 neurons are not necessary for egg-laying rhythmicity") of the Results, as well as a discussion of this in the third paragraph of the Discussion. In a nutshell, even though Howlader et al did not find a shortening when PDF neurons are ablated, they did find it in pdf01 flies.

(h) Figure 2 F, H As the authors mention in their Discussion on Page 16, lines 340-45, the manipulation of DN1p neurons might abolish the circadian rhythm in oogenesis as reported by Zhang et al, which is why they looked at this circuit driven by Clk4.1 neurons and comment that "The persistence of the rhythm of oviposition implies that it is not based on the availability of eggs but is instead an intrinsic property of the motor program". However, no change in fecundity is reported for either kir2.1 or perRNAi-based manipulations of these neurons, to help the reader understand if egg availability (at the level of egg formation) is playing any role in the downstream (and seemingly independent) act of egg laying. The authors should report if they see any change in total fecundity for either set of flies w.r.t their respective controls. Also, is the reduction in power seen with electrical silencing vs perRNAi expression of any relevance? Does the percentage of rhythmic flies change between these two manipulations?

In the line mentioned by the reviewer what we meant is that our results show that the rhythm of oviposition does not seem to be based in the rhythmic production of oocytes, which is not necessarily connected with the total number of eggs produced. We have modified the corresponding line in the paper, in order to avoid this misunderstanding. Regarding the "reduction in power" mentioned, it must be stressed that, in general, the height of the peak is correlated with the fraction of rhythmic individuals. The problem is that this fraction is a much more noisy output, and that is the reason why we have chosen to work with periodograms of averages.

(i) Figure 2 E and G, a loss of rhythmicity could also be due to a decrease in fecundity in the experimental lines. Since the number of eggs laid for each genotype is already known, can the authors show statistically relevant comparisons between the experimental lines and their respective controls? In this vein, can the averaged time series profiles also be provided for all the genotypes tested (as seen previously in Figure 1 A, C, G, I), perhaps in the supplementary?

We did not focus on fecundity in the present work. However, our observations do not seem to show any definite relationship with rhythmicity. We plan to address the issue of fecundity more systematically in a future work. The averaged time series profiles have now been added to the figure.

(j) Scatter plots showing the average period and SEM as seen in Figure 1 (F, L) would help in understanding if these manipulations have any effect on variation in the period of the egg-laying rhythm across flies. Particularly for pdf GAL4 > perRNAi flies which have a net shorter period, (but this might vary across the 34 flies tested).

We have added a Supplementary Figure (2S1) that shows that the shortening of oviposition period can be also observed at the individual level. We have also added a line commenting this in the corresponding subsection ("LNv and DN1 neurons are not necessary for egg-laying rhythmicity") of the Results, as well as a discussion of this in the third paragraph of the Discussion.

(k) Page 11 Figure 3B Does the presence of two peaks in the LS periodogram at a power > 0.2 indicate the presence of weakly rhythmic flies with both a short(20h) and a long(~27h) period component or either one? The short-period peak is nearly at p < 0.05 level of significance. So then, do most of the flies in MB122B GAL4 > perRNAi line show a weakly rhythmic shorter period?

(l) Figure 3D A similar peak is observed again at 20h (LS power > 0.2 and nearly at p < 0.05 significance level again) and a different longer one at (~30h) though this one is almost near 0.2 on the power scale. Given the consistency of this feature in both LNd manipulations, the authors should comment on whether this is driven by variation in periods detected or the presence of complex rhythms (splitting or change in period) in the oviposition time series for these lines.

(m) Figure 3 General scatter plots showing average period {plus minus} SEM could help explain the bimodality seen in the periodograms. Additionally indicating just how many flies are weakly rhythmic vs. strongly rhythmic can also help to illustrate how important the CRY+ LNds are to the oviposition rhythm's stability.

For these three comments (k, l and m), we note that the issue of bimodality has been addressed above, in our response to Weakness 9.

(o) Figure 4B Same as comments under Figure 1, what is the statistical test done to compare the proportions for these three genotypes?

As mentioned above, for the comparison of proportions we used the 'N-1' Chi-squared test. We have added a sentence detailing this at the end of the Statistical analysis section.

(p) Figure 4C Are all flies significantly rhythmic? The authors should also provide an averaged LS periodogram measure for each genotype, to help illustrate the difference in power between activity-rest and egg-laying rhythms.

Yes, the points represent periods of (significantly) rhythmic flies. This has been added to the caption, to avoid misunderstandings. The differences that arise when assessing rhythmicity in activity records vs. egg-laying records is addressed at length in the Supplementary Material (see e.g. Fig S1).

(q) Page 15 Figure 5 - general As the authors discuss the possible contribution of DN1ps to evening activity and control over oogenesis rhythm, investigating the connections of the few that are characterized in the connectome (or lack thereof) with the Oviposition neurons, can help illustrate the distinct role they play in the female Drosophila's reproductive rhythm.

This information was in the text and the Supplementary Tables. Lines 273-275 of the old manuscript read: "The full results are displayed in Supplementary Tables 2 and Table 3, but in short, we found that whereas there are no connections between LNV or DN1 neurons and oviposition neurons..."

(r) Minor: The dark shading of the circles depicting some of the clusters makes it difficult to read. Consider changing the colors or moving the names outside the circles.

Figure corrected.

(s) Line 38: The estimated number of clock neurons has been revised recently (<https://www.biorxiv.org/content/10.1101/2023.09.11.557222v2.article-info> [↗](#)).

Thank you for the reference. We have corrected the number of clock neurons in the Introduction of the new manuscript.

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